AURID

EU-Sovereign Microsoft 365 Replacement Platform

Product Vision · Platform Roadmap · Market Analysis · SWOT · Getting Started

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1. Executive Summary

Aurid is an EU-sovereign platform that enables European enterprises to replace Microsoft 365 entirely — built in phases, starting with Identity and Access Management and expanding to communications, mail, document editing, and file storage. Every layer runs on EU-owned infrastructure, authenticates through Aurid's identity core, and carries no US jurisdiction exposure.

Microsoft 365 is the operational backbone of European enterprise. It is deeply embedded, widely trusted, and genuinely capable. It is also a US product, operated under US law, subject to the CLOUD Act, and increasingly incompatible with European digital sovereignty obligations. As NIS2 compliance becomes mandatory and European institutions accelerate their move away from US-controlled infrastructure, the demand for a credible, complete, EU-sovereign alternative has never been clearer.

Aurid's approach is disciplined and phased. Phase 1 delivers a production-ready IAM platform — a managed, certified replacement for Microsoft Entra ID built on Keycloak and FreeIPA, with a complete Active Directory migration toolchain. This is the foundation on which everything else is built: every subsequent phase inherits a hardened, certified identity layer from day one. Phase 2 adds communications and mail — Element/Matrix replacing Teams, Tuta or Stalwart replacing Outlook and Exchange. Phase 3 completes the stack with Collabora Online replacing Microsoft 365 Office apps, and self-hosted object storage on Hetzner replacing OneDrive and SharePoint.

The phased model is deliberate. Phase 1 is fundable, saleable, and defensible as a standalone product. It generates the revenue and customer evidence needed to raise seed capital. That capital funds the certifications (ISO 27001, BSI C5) and go-to-market investment required to compete for enterprise accounts. Phase 2 and 3 expand the TAM and deepen customer lock-in with each successive phase. Customers who sign up for IAM are buying into a platform roadmap, not just an Entra ID replacement.

The company will be incorporated as a separate Danish ApS entity, distinct from J²-Consult ApS, with a clean cap table structured for venture investment. The founding team brings deep Keycloak and IAM engineering expertise, GxP and regulated-industry experience, and existing relationships in the pharmaceutical and life sciences sector.

2. The Problem

2.1 Identity is the New Perimeter — and It Lives in the US

Modern enterprise security is built on identity. Every application authentication, every access decision, every audit log flows through the Identity Provider (IdP). For the overwhelming majority of European enterprises, that IdP is Microsoft Entra ID — formerly Azure Active Directory.

Entra ID is a US product, operated by a US company, governed by US law. The CLOUD Act (Clarifying Lawful Overseas Use of Data Act) gives US authorities the legal right to compel Microsoft to hand over data stored on its servers — regardless of where those servers are physically located. A European Entra ID tenant in a Frankfurt datacenter is still, legally, subject to US government access.

This is not a theoretical risk. The EU-US Data Privacy Framework, which legitimises transatlantic data flows under GDPR, is politically fragile. The Trump administration's dismissal of Privacy and Civil Liberties Oversight Board members in early 2025 prompted formal concern from the European Parliament about the framework's adequacy. European enterprises are increasingly aware that their entire authentication infrastructure depends on a US company operating under US law.

2.2 NIS2 Is Creating Compliance Urgency

The NIS2 Directive, effective October 2024, significantly expands the scope and requirements for cybersecurity risk management across critical EU sectors — including healthcare, manufacturing, digital infrastructure, and financial services. Identity infrastructure is a primary attack vector and a primary audit focus.

NIS2 requires organisations to demonstrate control over their security measures, supply chain dependencies, and incident response capabilities. An identity platform operated by a foreign entity under foreign law creates a supply-chain dependency that is increasingly difficult to justify to regulators and auditors.

2.3 The Open Source Alternative Exists — But Nobody Has Packaged It

Keycloak is a mature, highly capable open-source IAM platform maintained by Red Hat. It supports OIDC, OAuth2, SAML, MFA, LDAP integration, and conditional access. FreeIPA provides enterprise directory services. The technology to replace Entra ID exists and is production-proven.

What does not exist is a managed service that packages this technology into an enterprise-ready product with the operational polish, support SLAs, compliance certifications, and clean admin experience that procurement and security teams require. This is the gap Aurid fills.

3. The Solution — Aurid

3.1 What Aurid Is

Aurid is a managed IAM platform delivered as a hybrid service. Customers connect their applications to Aurid as their Identity Provider in exactly the same way they would connect to Entra ID — via OIDC, SAML, or LDAP. The difference is where the data lives and who controls it.

3.2 Architecture

Aurid uses a hybrid control plane / data plane architecture, delivered on Kubernetes:

- Control Plane — operated by Aurid on Hetzner (Germany) or Scaleway (France) infrastructure. Handles tenant provisioning, billing, monitoring, the admin console, and policy management. EU jurisdiction throughout.
- Data Plane — deployed into the customer's own Kubernetes cluster, or into a dedicated Aurid-managed tenant cluster. Contains Keycloak (token issuance, OIDC/SAML), FreeIPA (directory), and audit log storage. The customer's identity data never leaves their chosen EU jurisdiction.
- OpenTofu Modules — Aurid provides production-ready OpenTofu (Terraform-compatible) modules for data plane deployment. Customers with Kubernetes competence can inspect and validate exactly what runs in their environment — a deliberate trust feature.
- Helm Charts — standard Kubernetes deployment units, auditable and reproducible.

This architecture gives customers the sovereignty guarantee — their identity data stays in their environment — while Aurid maintains the operational expertise, update cadence, and support capability that most organisations cannot maintain themselves.

3.3 Active Directory Replacement

The most significant migration challenge for organisations moving away from Entra ID is not the modern OIDC/SAML layer — it is the legacy Active Directory dependency. AD provides directory services (LDAP), Kerberos authentication for Windows workstations and servers, Group Policy Objects (GPO) for device and user configuration, and DNS integration that underpins the entire domain infrastructure.

Aurid replaces AD with FreeIPA — a mature, enterprise-grade open-source directory platform that provides native LDAP, Kerberos, DNS, and certificate authority services. FreeIPA is production-proven in large-scale deployments and is the upstream project behind Red Hat Identity Management.

To make this migration tractable for organisations without deep Linux directory expertise, Aurid ships aurid-migrate as part of the MVP — a dedicated migration toolchain that automates the extraction, transformation, and import of AD data into FreeIPA. See Appendix A for a detailed technical walkthrough of the migration process and tool architecture.

3.3 EU Infrastructure

Aurid operates exclusively on EU-headquartered, EU-owned cloud infrastructure:

- Hetzner Online GmbH — German-owned, German-operated. Excellent price/performance, strong sovereignty credentials, no US ownership.
- Scaleway (Iliad Group) — French-owned. Strong EU enterprise credibility, particularly for French-regulated industries.

Customers can select their preferred control plane location. Data plane deployment is entirely within the customer's own infrastructure or chosen EU region, giving full data residency control.

3.4 The Admin Experience

Keycloak's raw administration interface is functional but unloved. Aurid wraps the platform in a clean, modern administration console purpose-built for enterprise IT administrators — not identity engineers. This is a primary product differentiator. If an IT administrator can manage Entra ID, they can manage Aurid without retraining.

3.5 GxP and Regulated Industry Readiness

Aurid's audit logging is designed from the ground up for regulated industries. Immutable, tamper-evident audit trails stored on EU infrastructure satisfy GxP (21 CFR Part 11 / Annex 11) requirements, making Aurid immediately credible to pharmaceutical and life sciences organisations — a natural early market given the founding team's background.

4. Microsoft Entra ID Capability Coverage

The table below maps Entra ID's core capabilities to Aurid's planned coverage. MVP capabilities are targeted for the initial product release. Phase 2 capabilities follow the first funding round.

Entra ID Capability	Aurid Coverage	Notes
OIDC / OAuth 2.0 (modern SSO)	✓ MVP	Keycloak core — full standards compliance
SAML 2.0 (legacy enterprise apps)	✓ MVP	Keycloak native support
Multi-Factor Authentication (TOTP, WebAuthn / FIDO2)	✓ MVP	Hardware key support included
Directory services (LDAP / AD-compatible)	✓ MVP	FreeIPA / OpenLDAP integration layer
User lifecycle management (provisioning / deprovisioning)	✓ MVP	SCIM 2.0 protocol support
AD Migration Tool (aurid-migrate)	✓ MVP	LDIF export, user/group/GPO mapping — see Appendix A
Single Sign-On across SaaS applications	✓ MVP	Pre-built connectors for top 30 EU SaaS apps
Group and role management (RBAC)	✓ MVP	Fine-grained attribute-based access control
Audit logging (tamper-evident)	✓ MVP	GxP-grade immutable audit trail on EU storage
Basic conditional access (location, IP, time)	✓ MVP	Policy engine in control plane
Password policies and self-service reset	✓ MVP	Configurable per tenant
B2B federation (guest / partner access)	◑ Phase 2	OIDC federation with external IdPs
Device trust / MDM integration	◑ Phase 2	Fleet integration for device posture checks
Risk-based / adaptive conditional access	◑ Phase 2	Behaviour analytics layer
Privileged Identity Management (PIM)	◑ Phase 2	Just-in-time admin access elevation
Identity Governance (access reviews, entitlement)	✗ Roadmap	Post-Series A priority
Verified ID / Verifiable Credentials	✗ Roadmap	Decentralised identity — future capability

Coverage legend: ✓ MVP = available at initial product launch ◑ Phase 2 = post-seed funding ✗ Roadmap = longer-term vision

5. Market Analysis

5.1 Market Size

The global Identity and Access Management market is valued at approximately USD 20–25 billion in 2025, growing at a CAGR of 13–15% through 2030. The European segment represents roughly 25–30% of global spend — approximately USD 5–7 billion — with enterprise CIAM and workforce IAM as the dominant categories.

Aurid's initial addressable market is European enterprises of 200–5,000 employees currently running Entra ID P1 or P2, with sovereignty concerns or NIS2 compliance obligations. We estimate 15,000–25,000 such organisations across the EU and Nordic markets. At a target price of €5/user/month and an average of 500 users, this represents a serviceable addressable market of approximately €450M ARR.

5.2 Competitive Landscape

Competitor	HQ / Sovereignty	Price Signal	Key Gap vs Aurid
Microsoft Entra ID P2	US — CLOUD Act exposure	€9–12/user/mo	No EU sovereignty; US legal jurisdiction
Okta Workforce Identity	US — CLOUD Act exposure	€12–18/user/mo	US company; premium pricing; no sovereignty story
Ping Identity	US (Thales-owned)	Enterprise custom	Complex; expensive; no managed EU-sovereign option
Zitadel	Switzerland (open-source)	Self-host or cloud	No managed service; requires internal ops team
Authentik	Open-source community	Self-host only	No enterprise support; no certifications
Keycloak (raw)	Red Hat / IBM (US)	Self-host only	Requires significant ops expertise; not a product
Aurid (us)	Denmark / EU — sovereign	€4–7/user/mo target	Managed, certified, EU-sovereign — the gap we fill

5.3 Market Tailwinds

- NIS2 Directive enforcement accelerating IAM investment and supply-chain security scrutiny across EU enterprise
- EU-US Data Privacy Framework fragility raising boardroom-level concern about US-jurisdiction data dependencies
- Trump administration actions — PCLOB dismissals, CLOUD Act enforcement — increasing European sovereignty sentiment
- Microsoft price increases on M365 / Entra creating switching motivation on cost grounds alone

- Growing awareness that 'EU datacenter' does not equal 'EU jurisdiction' when the operator is a US company
- Pharmaceutical and life sciences sector under increasing regulatory scrutiny for data sovereignty and GxP compliance

5.4 Target Customer Profile — Early Adopters

Initial target customers share a combination of sovereignty awareness and practical switching motivation:

- European mid-market enterprises (200–2,000 employees) with a CISO or DPO actively engaging with NIS2 obligations
- Pharmaceutical and life sciences companies with existing GxP compliance infrastructure and data sovereignty concerns
- Public sector adjacent organisations (NGOs, research institutions, EU-funded entities) with explicit data sovereignty mandates
- Professional services firms (law, audit, consulting) handling sensitive EU client data under GDPR obligations
- Nordic enterprises with strong cultural alignment to privacy and digital sovereignty

6. SWOT Analysis

Strengths

- Deep Keycloak / IAM engineering expertise
- GxP & regulated-industry pedigree from pharma work
- Hybrid Kubernetes-native architecture — EU data residency by design
- No US CLOUD Act exposure on data plane
- First-mover in EU-sovereign managed identity space

Weaknesses

- Brand unknown — zero market presence at launch
- Small team; limited capacity for concurrent deals
- No certifications at outset (ISO 27001, NIS2)
- Enterprise sales cycles are long without existing relationships
- Limited budget to fund compliance and go-to-market simultaneously

Opportunities

- NIS2 Directive creating urgent compliance demand across EU
- EU-US Data Privacy Framework politically fragile — sovereignty urgency rising
- Trump administration actions accelerating European tech sovereignty sentiment
- Entra ID pricing increases creating cost-motivated switching interest
- No credible EU-sovereign IAM managed service exists today
- Pharma sector deeply Microsoft-dependent but increasingly sovereignty-aware

Threats

- Zitadel and Authentik could productize faster if well-funded
- Okta / Ping could launch EU-sovereign offerings
- Microsoft could add EU-sovereign Entra tier
- ISO 27001 gap creates procurement blockers pre-certification
- Key-person risk with small founding team
- EU-US data framework stabilising could reduce urgency

6.1 Strategic Implications

The SWOT analysis points to a clear strategic path: move fast enough to establish market presence before a well-funded competitor occupies the space, use the pharma/regulated-industry wedge to acquire credible early customers whose procurement processes are rigorous enough to validate the product, and treat ISO 27001 certification as the single most important non-product investment in year one post-funding.

The key risk to manage is the window between product availability and certification. Design partners who commit to pilots before certification is complete — in exchange for pricing advantages and roadmap influence — are essential to bridge this gap and provide the investor narrative.

7. Getting Started — Execution Plan

7.1 Phased Platform Roadmap

Aurid is built in three major phases, each fundable and saleable as a standalone proposition. The phasing is non-negotiable — IAM must be production-hardened and certified before communications is added, and communications must be proven before document editing and file storage. Every phase inherits the certified identity layer of the phase before it.

Phase	Timeline	Key Deliverables
Phase 0 Foundation	Now – Month 2	Incorporate Aurid ApS. Define founding team. Acquire aurid.com / aurid.eu / aurid.io domains. Define reference architecture. Provision Hetzner and Scaleway control plane infrastructure. Set up GitOps pipeline.
Phase 1A IAM MVP	Month 2 – 6	Keycloak core, FreeIPA directory, OIDC/SAML 2.0/MFA, admin console v1, tamper-evident audit logging. Helm charts and OpenTofu modules for data plane deployment. aurid-migrate AD migration toolchain.
Phase 1B IAM Market	Month 6 – 14	2–3 design partners onboarded. ISO 27001 gap assessment and remediation. NIS2 alignment documentation. Penetration test report published. Seed round raise (€1–2M). Commercial pricing launched.
Phase 1C IAM Certified	Month 14 – 20	ISO 27001 certification achieved. BSI C5 assessment initiated. Go-to-market hire. Active sales to EU enterprise. B2B federation and device trust (Phase 2 IAM features). Nordic and DACH expansion begins.
Phase 2A Comms MVP	Month 18 – 24	Element/Matrix deployment (ESS Pro) as managed service in customer data plane. Aurid OIDC federation with Element out of the box. Tuta Business integration (referral/partner) or Stalwart Mail self-hosted option. Unified admin console covering IAM + comms.
Phase 2B Comms Market	Month 24 – 30	Teams + Outlook replacement package commercially available. Joint go-to-market with Element. First full 'leave Microsoft' customers live. Series A preparation. Expand comms certification coverage.
Phase 3A Docs MVP	Month 28 – 36	Collabora Online deployed as managed service in customer data plane. SFTPGo + Hetzner Object Storage as self-owned file storage layer (OneDrive/SharePoint replacement). WebDAV native OS mounting. Aurid SSO across all three pillars.
Phase 3B Full Platform	Month 36+	Complete EU-sovereign Microsoft 365 replacement commercially available: IAM + Communications + Mail + Document Editing + File Storage. Series A close. BSI C5 certification. Desktop sync client (Phase 2 file storage feature).

Phase boundaries are indicative. The transition from Phase 1C to Phase 2A begins once ISO 27001 is achieved and the first commercial customers are live — not on a fixed calendar date. The discipline is: each phase must be genuinely complete before the next begins.

7.2 Immediate Actions (Next 60 Days)

Legal & Corporate

- Incorporate Aurid ApS as a separate legal entity from J²-Consult ApS
- Establish holding structure if VC investment is anticipated — consult startup-focused lawyer before filing
- Acquire aurid.com domain (currently listed for sale on MarkUpgrade) plus aurid.eu and aurid.io
- Draft co-founder / team agreements if additional founders are brought in

Technical Foundation

- Define reference architecture — control plane topology, data plane deployment model, network boundaries
- Provision Hetzner and Scaleway accounts; establish Kubernetes clusters for control plane
- Set up GitOps pipeline (Flux or ArgoCD) for control plane deployment
- Begin Keycloak configuration: realm structure, client templates, MFA policies
- Design admin console wireframes — this is the primary UX differentiator, invest early
- Begin aurid-migrate discovery module — the AD inventory report is a high-value sales tool even before migration capability is complete

Market & Commercial

- Identify 3–5 design partner candidates from existing network (J²-Consult relationships, pharma contacts)
- Develop a one-page capability brief for design partner conversations covering Phase 1 IAM + full platform roadmap
- Begin security and architecture documentation — penetration test scope, threat model, architecture whitepaper
- Initiate ISO 27001 gap assessment — understand the delta before pitching investors
- Make initial contact with Element and Collabora as future technology partners — no commitment needed now, just relationship building

7.3 Team

The founding team of 2–3 people should cover three distinct capability areas:

- Technical Architecture & Engineering (founder) — Keycloak, Kubernetes, IAM platform, infrastructure. This is the core IP of the product.
- Go-to-Market & Enterprise Sales — the hardest hire for a technical founding team and the most critical. Ideally someone with existing relationships in EU enterprise IT or the compliance/security buyer community.
- Security & Compliance — ideally someone who has previously led an ISO 27001 certification. The process is long and painful to navigate for the first time; experienced guidance is worth its cost.

7.4 Funding Strategy

The funding sequence maps directly to the platform phases, with each round de-risked by the completion of the previous phase:

- Bootstrap (Phase 0–1A) — from J²-Consult revenue and personal capital. Keep burn low; focus on MVP and design partners. Target: working product, 2 design partners, ISO 27001 gap assessment in hand.
- Seed Round €1–2M (Phase 1B) — raised once design partners are live. Narrative: proven IAM product, regulated-industry validation, clear certification roadmap, full platform vision. Capital funds ISO 27001, BSI C5, go-to-market hire, and Phase 2 build.
- Series A (Phase 2B–3A) — raised once communications stack is live and first 'leave Microsoft' customers are on contract. Narrative: full platform, certified, growing ARR, expanding TAM. Capital funds Phase 3 build, DACH/Benelux expansion, second go-to-market hire.

Target investor profiles: Northzone (Nordic, enterprise software), Seed Capital Denmark (early stage Danish), NATO Innovation Fund (sovereignty/defence adjacency), Eurazeo (French, EU sovereign tech), Sovereign Tech Fund Germany (open source infrastructure grants — non-dilutive, worth pursuing early).

7.5 Pricing Model

Pricing evolves with the platform. Phase 1 pricing is IAM-only and positions below Entra ID P2. Phase 2 and 3 introduce bundle pricing that makes the full platform significantly cheaper than the Microsoft 365 equivalent while maintaining healthy margins.

- IAM Starter — up to 500 users, shared control plane, standard SLA: €4/user/month
- IAM Business — up to 5,000 users, dedicated tenant, enhanced SLA, GxP audit logging: €6/user/month
- IAM Enterprise — unlimited users, dedicated data plane in customer infrastructure, custom SLA, compliance package: custom pricing
- Platform Bundle (Phase 2+) — IAM + Communications + Mail: €10–12/user/month. Compared to Microsoft 365 Business Premium at €22/user/month, this represents 45–55% cost reduction with full sovereignty.
- Full Platform (Phase 3+) — all layers including document editing and file storage: €14–16/user/month. Microsoft 365 equivalent would be €22–28/user/month.

Early adopter customers who sign contracts during Phase 1 receive locked pricing for all subsequent phases — this is the commercial incentive for design partners to commit early and for Phase 1 customers to stay on the platform as it grows.

8. Partner Ecosystem & Platform Stack

8.1 The Full Stack

Aurid is the identity layer that every other component in the platform authenticates through. The components below are either managed by Aurid as part of the service (running in the customer data plane), owned Aurid-built capabilities (file storage), or partner integrations (Tuta, Element). In every case, the data stays in the EU and the identity stays in Aurid.

Platform Layer	Component	Replaces	Phase	Notes

Identity & IAM	**Aurid (Keycloak + FreeIPA)**	Entra ID + AD	**Phase 1**	Core product. Everything else authenticates through this.
Identity & IAM	**aurid-migrate**	AD migration	**Phase 1**	CLI + UI toolchain for AD-to-FreeIPA migration. Ships with MVP.
Communications	**Element / Matrix (ESS Pro)**	Microsoft Teams	**Phase 2**	OIDC-native federation with Aurid. Self-hosted in customer data plane.
Mail & Calendar	**Tuta Business**	Outlook / Exchange	**Phase 2**	German E2E encrypted SaaS. Partner/referral model. No IMAP.
Mail & Calendar	**Stalwart Mail**	Outlook / Exchange	**Phase 2**	Self-hosted alternative. Full IMAP/SMTP. LDAP-native. OIDC SSO.
Document Editing	**Collabora Online**	Microsoft 365 Office apps	**Phase 3**	LibreOffice-based. Real-time collaboration. WOPI protocol. Self-hosted.
File Storage	**SFTPGo + Hetzner Object Storage**	OneDrive / SharePoint	**Phase 3**	Aurid-owned capability. S3-compatible. WebDAV for OS-native mounting.
File Storage	**MinIO (in-cluster)**	OneDrive / SharePoint	**Phase 3**	Optional in-cluster object storage for customers requiring full data plane ownership.

8.2 Phase 1 — Identity & IAM

The IAM layer is Aurid's core product and the foundation for everything else. Keycloak handles all modern protocol surfaces (OIDC, OAuth2, SAML 2.0). FreeIPA provides the enterprise directory (LDAP, Kerberos, DNS, CA). aurid-migrate handles the transition from Active Directory. The admin console wraps all of this in an interface that an IT administrator can operate without identity engineering expertise.

This phase is complete when Aurid can replace Entra ID and AD for any European enterprise, certified to ISO 27001, with a reference customer in a regulated industry.

8.3 Phase 2 — Communications & Mail

Element / Matrix — Teams Replacement

Element is the leading EU-sovereign Teams alternative, built on the open Matrix protocol. It is deployed as Element Server Suite (ESS Pro) in the customer data plane — a Kubernetes-native deployment that Aurid manages. Aurid Keycloak is the OIDC provider; Element federates natively without any custom integration work.

Matrix is gaining serious institutional momentum: France has 600,000 government officials on Tchap (a Matrix deployment), Germany's Bundeswehr deployed it to 100,000 military personnel, and the European Commission is actively piloting it as a Teams replacement. These are the reference customers Aurid's target market reads and trusts.

Mail & Calendar — Tuta or Stalwart

Aurid offers two mail options to match customer requirements:

- Tuta Business — German SaaS, end-to-end encrypted by default, calendar included, no IMAP (encrypted architecture precludes it). Best for customers prioritising encryption purity over client flexibility. Aurid refers customers and ensures SSO compatibility; Tuta operates the service.
- Stalwart Mail — self-hosted in the customer data plane. Full IMAP/SMTP/JMAP, LDAP-native (connects to FreeIPA directly), OIDC SSO via Keycloak. Best for customers requiring standard mail clients, full operational control, and deep directory integration. Aurid manages the deployment.

Both options are presented to customers during Phase 2 onboarding. The choice is documented in the service agreement.

8.4 Phase 3 — Document Editing & File Storage

Collabora Online — Office Apps Replacement

Collabora Online is the enterprise web-based edition of LibreOffice, developed by Collabora Productivity (Cambridge, UK) — the largest contributor to the LibreOffice codebase. It provides Writer (Word), Calc (Excel), and Impress (PowerPoint) running in the browser with real-time multi-user collaboration, full .docx/.xlsx/.pptx compatibility, and self-hosted deployment via Docker or Kubernetes.

Collabora Online connects to file storage via the WOPI protocol — the same protocol Microsoft 365 uses internally. It works natively against any WebDAV or S3-compatible storage backend, including Aurid’s file storage layer, without any custom integration.

File Storage — Aurid-Owned Capability

Unlike the other platform layers, file storage is not a third-party product — it is built and operated by Aurid on infrastructure Aurid controls. This is a deliberate sovereignty decision: file storage is where the most sensitive enterprise data lives, and pointing customers at another vendor’s platform creates an unwanted dependency.

The file storage architecture:

- Hetzner Object Storage as the storage backend — German-owned, S3-compatible, €0.0085/GB. For customers requiring full data plane ownership, MinIO is deployed in-cluster instead.
- SFTPGo as the access layer — a single Go binary running in Kubernetes that exposes WebDAV (native OS mounting on Windows and macOS without any client software), SFTP, and S3 presigned URLs. Supports per-user and per-group storage quotas. OIDC/OAuth2 authentication connects natively to Aurid Keycloak.
- Collabora Online connects to SFTPGo via WOPI for in-browser document editing.
- Element connects to object storage for media and file attachments.

- A lightweight desktop sync client (Tauri-based, wrapping rclone) is a Phase 3 follow-on feature for customers who require transparent Windows Explorer / Mac Finder integration.

This architecture gives customers a complete OneDrive and SharePoint replacement with zero third-party SaaS dependencies, running entirely within their chosen EU jurisdiction, authenticated through Aurid.

8.5 What Aurid Does Not Build

Platform discipline requires knowing what not to build. The following are explicitly out of scope regardless of customer requests:

- Video conferencing infrastructure — Element Call (built on LiveKit and MatrixRTC) handles this natively as part of the Element deployment. Aurid does not build or operate separate video infrastructure.
- Email hosting — Tuta operates their own infrastructure; Stalwart Mail runs in the customer data plane. Aurid does not operate a shared email hosting service.
- A proprietary office suite — Collabora Online is the office layer. Aurid does not build document editing software.
- A consumer product — Aurid is enterprise-only. No freemium tier, no self-service signup, no consumer marketing.

9. Summary

Aurid is not just another IAM product. It is the identity foundation for a complete, EU-sovereign alternative to Microsoft 365 — built in phases, with the discipline to do each phase properly before moving to the next.

Phase 1 — IAM — is where it starts. A production-hardened, certified, managed replacement for Entra ID and Active Directory, with a migration toolchain that makes the transition tractable. This phase is fundable, saleable, and defensible on its own. It generates the revenue, customer evidence, and certifications needed to raise seed capital and build Phase 2.

Phase 2 adds communications and mail — Element replacing Teams, Tuta or Stalwart replacing Outlook. This is when the 'leave Microsoft entirely' conversation becomes possible with customers. The TAM expands from IAM buyers to the full M365 replacement market.

Phase 3 completes the platform — Collabora Online for document editing, Aurid-owned file storage on Hetzner replacing OneDrive and SharePoint. At this point Aurid is a complete Microsoft 365 replacement at 40–50% lower cost, with full EU sovereignty, no US jurisdiction exposure, and a single identity layer running everything.

The timing is right. NIS2 is creating urgency. The Trump administration's actions are creating anxiety. The open source components are mature. The institutional deployments — French government, German Bundeswehr, European Commission — are creating the reference customer credibility that enterprise procurement needs. What has been missing is a team that can package all of it into a managed, certified, enterprise-grade service. Aurid is that team.

Aurid ApS — The EU-sovereign platform that replaces Microsoft 365. Built in phases. Built to last.

Appendix A: Active Directory Replacement — Architecture & Migration

A.1 Why AD Replacement is the Hard Problem

When organisations evaluate leaving Microsoft's identity stack, the conversation often starts with Entra ID's cloud layer — the OIDC endpoints, the Conditional Access policies, the SSO integrations. These are well-understood standards and replacing them with Keycloak is relatively straightforward. The harder problem is what sits beneath: on-premises Active Directory.

Active Directory has been the backbone of enterprise Windows environments since Windows 2000. It does far more than store usernames and passwords. It provides:

- LDAP directory services — the authoritative store for users, groups, computers, and organisational units
- Kerberos authentication — ticket-based authentication for Windows domain-joined machines, file servers, printers, and legacy applications
- Group Policy Objects (GPO) — centralised configuration management for Windows workstations and servers (desktop settings, software deployment, security baselines, mapped drives)
- DNS integration — AD-integrated DNS zones that underpin domain name resolution across the entire enterprise network
- Certificate Services — enterprise CA for internal TLS certificates, code signing, and smart card authentication
- Trust relationships — cross-domain and cross-forest trusts enabling authentication across organisational boundaries

Replacing each of these capabilities requires a deliberate technical strategy. Aurid's approach is to use FreeIPA as the directory and Kerberos core, supplement it with purpose-built tooling for GPO migration, and provide the aurid-migrate toolchain to automate the data extraction and transformation process.

A.2 The Aurid Directory Stack

FreeIPA as the AD Replacement

FreeIPA is the upstream open-source project for Red Hat Identity Management. It integrates MIT Kerberos, 389 Directory Server (LDAP), Dogtag Certificate System, and BIND DNS into a single, coherent identity platform. In a Kubernetes deployment, Aurid runs FreeIPA in dedicated pods within the customer data plane.

FreeIPA provides native equivalents for the core AD services:

AD Capability	FreeIPA Equivalent	Notes
LDAP Directory (AD DS)	389 Directory Server	Full LDAP v3 compliance; RFC 2307 schema for Linux attrs

Kerberos (NTLM fallback)	MIT Kerberos 5	Native Kerberos; SSSD on Linux clients; Samba for Windows
AD-integrated DNS	BIND + FreeIPA DNS plugin	Dynamic DNS updates; SRV records auto-managed
Certificate Services (AD CS)	Dogtag Certificate System	Enterprise CA; auto-enrollment via ACME or SCEP
Group Policy Objects (GPO)	Ansible + aurid-policy engine	GPO-to-Ansible translation — see section A.4
Windows domain join	Samba 4 / SSSD	Windows machines join FreeIPA realm via Samba 4 AD compatibility layer
Trust relationships	FreeIPA cross-realm trusts	AD-to-FreeIPA trust during migration; FreeIPA-to-FreeIPA post

Keycloak as the Modern Identity Broker

FreeIPA handles the directory and Kerberos layer. Keycloak sits above it as the modern identity broker — federating against FreeIPA via LDAP and Kerberos, and presenting OIDC, OAuth2, and SAML interfaces to applications. This two-layer architecture mirrors how Entra ID itself works: AD DS underneath, modern protocols on top.

The advantage of this separation is that applications consume standard protocols from Keycloak and never need to know whether the directory underneath is FreeIPA, another LDAP source, or a future replacement. The directory layer is operationally invisible to application developers.

A.3 aurid-migrate: The Migration Toolchain

aurid-migrate is a CLI tool and optional web UI shipped as part of the Aurid MVP. It automates the most labour-intensive parts of an AD-to-FreeIPA migration. The tool is designed to be run by an IT administrator with AD access — not an identity engineer — and produces a fully documented migration package that Aurid's onboarding team reviews before any changes are made to production.

Phase 1: Discovery and Inventory

aurid-migrate connects to the source AD environment (read-only service account) and produces a comprehensive inventory report:

- User objects — count, attribute completeness, disabled accounts, stale accounts (no login >90 days), password policy assignments
- Group objects — security groups vs distribution groups, nesting depth, empty groups, groups with no members
- Organisational Units — OU tree structure, GPO links per OU, object counts
- Computer objects — domain-joined workstations and servers, OS versions, last seen dates
- Service accounts — accounts with SPNs (Service Principal Names), delegation settings, password age

- GPO inventory — all policies enumerated, settings categories, linked OUs, WMI filters
- Trust relationships — existing domain and forest trusts
- Schema extensions — custom AD attributes that require mapping decisions

The inventory report is the first deliverable to the customer — often the most complete view of their AD environment they have ever had. It serves as the migration scope document and identifies complexity that affects effort and timeline estimates.

Phase 2: Export and Transformation

aurid-migrate performs a staged export and transformation of AD data into FreeIPA-compatible format:

- LDIF generation — users, groups, and OUs exported as LDIF (LDAP Data Interchange Format), with attribute mapping from AD schema to RFC 2307 / FreeIPA schema. Handles common mismatches: sAMAccountName → uid, userPrincipalName → mail, memberOf → member.
- Password hash handling — AD password hashes use NTLM format, which FreeIPA cannot import directly. aurid-migrate generates a one-time password reset notification workflow so users set new passwords on first login to the FreeIPA-backed environment, without requiring IT to reset passwords individually.
- Group nesting flattening — deeply nested AD groups are optionally flattened to a configurable depth, reducing complexity in the target environment.
- SPN mapping — Service Principal Names are translated to FreeIPA service principals, preserving Kerberos delegation for applications that require it.
- OU structure mapping — AD OU hierarchy is reproduced in FreeIPA, or optionally rationalised into a flatter structure during migration.
- Conflict detection — identifies duplicate sAMAccountNames, malformed UPNs, illegal characters, and attribute length violations before import.

Phase 3: Validation and Dry Run

Before any data is written to the target FreeIPA environment, aurid-migrate performs a full dry-run import against a staging FreeIPA instance. The dry run:

- Validates all LDIF against the FreeIPA schema and reports errors with line-level detail
- Resolves group membership references and reports dangling members
- Produces a diff report: objects that will be created, modified, or skipped
- Generates a rollback plan — the LDIF required to remove all imported objects if the migration needs to be reversed

The dry-run report is reviewed by the customer’s IT team and Aurid’s onboarding engineer before proceeding to production import.

Phase 4: Production Import

Production import is performed in a controlled maintenance window. aurid-migrate:

- Imports users, groups, OUs, and service principals in dependency order (OUs first, then users, then groups)

- Establishes an AD-to-FreeIPA trust relationship, allowing both directories to coexist during a parallel-run period
- Configures Keycloak federation against FreeIPA via LDAP, enabling immediate SSO testing against the new directory
- Logs every operation to the Aurid audit trail with before/after state, operator identity, and timestamp

Phase 5: Parallel Run and Cutover

Aurid strongly recommends a parallel-run period of 2–4 weeks before decommissioning AD. During parallel run:

- Both AD and FreeIPA are live; the AD-to-FreeIPA trust allows authentication against either
- Applications are migrated to Keycloak OIDC/SAML one by one, with rollback possible per application
- Windows workstations remain AD-joined; Linux clients migrate to FreeIPA/SSSD
- aurid-migrate sync daemon runs continuously, propagating new users and group changes from AD to FreeIPA to keep directories in sync during the transition

Cutover is the moment AD is demoted and FreeIPA becomes authoritative. At this point Windows workstations are re-joined to the FreeIPA realm via Samba 4, and the AD trust is removed.

A.4 GPO Migration — The Awkward Problem

Group Policy Objects are the most operationally complex part of an AD migration. GPOs are a Windows-native configuration management mechanism with no direct equivalent in the Linux/open-source world. Aurid’s approach is pragmatic rather than attempting a like-for-like replacement.

GPO Categories and Their Replacements

GPO Category	Aurid Replacement	Detail
Security settings (password policy, account lockout, audit)	FreeIPA password policy + aurid-policy	Native FreeIPA per-group password policies; audit settings via aurid-policy Ansible module
Software deployment	Ansible + package manager	Ansible playbooks replace GPO-based MSI deployment; integrates with existing MDM if present
Drive mappings / network shares	Ansible login scripts	Mapped to user/group attributes in FreeIPA; applied via SSSD or login scripts
Desktop / wallpaper / UI settings	MDM (Jamf / Fleet) or Ansible	OS-level settings migrated to MDM for managed devices; Ansible for servers
Windows Firewall rules	Ansible Windows modules	ansible.windows.win_firewall_rule; can be driven from FreeIPA group membership

Certificate auto-enrollment	Dogtag CA + ACME / SCEP	Dogtag provides ACME endpoint; clients enrol via certbot or OS-native ACME support
Scripts (logon/logoff/startup/shutdown)	FreeIPA HBAC + Ansible	Host-Based Access Control in FreeIPA replaces many script use cases; complex scripts ported to Ansible

aurid-migrate includes a GPO analyser that categorises all GPOs in the source environment by the above taxonomy and produces a migration effort estimate. Simple security and password policy GPOs are automatically translated to FreeIPA equivalents. Complex GPOs (software deployment, custom registry settings) are flagged for manual review and converted to Ansible playbooks with Aurid's onboarding assistance.

GPO migration is the part of an AD replacement most likely to require professional services hours. Aurid's commercial model includes a fixed-price GPO migration service for customers with complex policy environments.

A.5 Windows Client Integration

Windows workstations are typically the last piece of an AD migration to move. Aurid supports two approaches depending on the customer's Windows estate:

Option 1: Samba 4 AD Compatibility (Recommended for Windows-heavy environments)

Samba 4 implements the AD protocols natively — Kerberos, LDAP, DNS, and the MS-specific RPC interfaces that Windows clients use for domain join. FreeIPA can be configured to use Samba 4 as its AD compatibility layer, allowing Windows workstations to domain-join to a FreeIPA realm as if it were an AD domain.

This approach requires no changes to Windows clients. Group Policy processing continues to work for supported GPO categories. The migration is transparent to end users — they log in the same way, with the same credentials, from the same Windows machines.

Option 2: Azure AD Join Replacement via Intune-Alternative MDM

For organisations that have already moved workstations to Azure AD Join (Entra Join) rather than on-premises AD, Aurid supports a migration path via MDM-only management. Workstations are unenrolled from Intune and Entra Join, then enrolled into a FreeIPA-backed MDM — either Fleet (open-source, cross-platform) or Jamf for macOS-heavy environments — with identity provided by Keycloak.

This option involves more end-user visible change during cutover and is recommended only for organisations already operating in a modern device management posture without heavy GPO dependency.

A.6 Migration Timeline and Effort Estimates

Organisation Size	Estimated Duration	Professional Services	Key Complexity Drivers
Small (50–200 users)	4–6 weeks	5–10 days	Simple AD structure; few GPOs; limited legacy app dependencies
Mid-market (200–1,000 users)	8–16 weeks	15–25 days	Multiple OUs; moderate GPO complexity; mixed Windows/Linux estate
Enterprise (1,000–5,000 users)	16–26 weeks	30–60 days	Multiple domains/forests; complex GPO estate; legacy Kerberos applications; extended parallel run

These estimates assume a well-documented source AD environment and a customer IT team available to support the migration. aurid-migrate's automated tooling reduces professional services effort by approximately 40–60% compared to a fully manual migration.

A.7 What aurid-migrate Does NOT Handle

Aurid is transparent about the boundaries of automated migration. The following require manual assessment and are out of scope for the migration toolchain:

- Exchange / Exchange Online — mail infrastructure migration is a separate workstream. Aurid provides directory integration with open-source mail platforms (Nextcloud Mail, Stalwart) but does not migrate mailboxes.
- SharePoint / OneDrive — document storage migration is out of scope. Aurid integrates with Nextcloud for file storage but does not perform data migration.
- Custom AD schema extensions — organisations with heavily customised AD schemas (common in pharma and manufacturing with HR system integrations) require schema analysis and custom attribute mapping before migration.
- Legacy applications using NTLM authentication — NTLM is a Microsoft-proprietary protocol with no open-source equivalent. Applications relying on NTLM must be updated to use Kerberos or modern protocols before migration. aurid-migrate flags all NTLM-dependent SPNs in the discovery phase.
- Azure AD / Entra-specific features — Conditional Access policies, Entra-specific app registrations, and Microsoft-specific B2B federation are not migrated. These are replaced by Keycloak equivalents with equivalent functionality, but require reconfiguration rather than automated migration.

The discovery phase of aurid-migrate specifically surfaces all of these blockers so that scope is fully understood before commercial commitments are made.

A.8 Appendix Summary

The AD replacement problem is real, complex, and the primary reason most organisations do not attempt to leave the Microsoft identity stack. Aurid's approach is to make it tractable, not to pretend it is easy.

aurid-migrate ships as part of the MVP because it is the product feature that unlocks the market. Without a credible migration path, Aurid is an interesting greenfield option only. With it, Aurid is a viable replacement for organisations that already have an AD investment and want to move.

The commercial model around migration — tooling included in the subscription, professional services for complex environments — creates an additional revenue stream and deepens customer relationships during onboarding. Customers who have completed a successful AD migration with Aurid's assistance are highly unlikely to churn.